Project Specifications

[Team GitHub Repository(Click Here)]

1. Project Topic

Cross-Domain Sentiment Analysis: Evaluating Model Generalization Across Distinct Industries. This project investigates the engineering feasibility of domain adaptation in machine learning. It focuses on evaluating the performance degradation and adaptability of sentiment analysis models when trained on one specific commercial domain (source) and deployed on an entirely unseen commercial domain (target) without retraining.

2. Problem Statement & Research Question

Problem Statement: E-commerce platforms and digital aggregators frequently struggle to maintain high-accuracy sentiment classification when applying a model trained on one product category to a new, distinctly different category. Building isolated models for every commercial domain is computationally inefficient and expensive.
Research Question: To what extent does a machine learning sentiment classifier's accuracy and F1-score degrade when trained on a source domain (e.g., Electronics reviews) and evaluated on an unseen target domain (e.g., Movies or Restaurant reviews), and can a custom hybrid model utilizing transfer learning mitigate this degradation?

3. Project Objectives

To systematically answer the research question using the CDIO (Concept, Design, Implement, Operate) framework, the team will execute the following objectives:

- **Establish Baseline Metrics:** Train and evaluate a standard neural network exclusively on the source domain, recording baseline inference metrics on the unseen target domains.
- **Implement Transfer Learning:** Fine-tune a pre-trained Transformer architecture on the source domain to evaluate the inherent adaptability of deep semantic embeddings.
- **Engineer a Custom Hybrid Architecture:** Develop a model that concatenates the deep semantic embeddings extracted from the pre-trained transformer with handcrafted, statistical linguistic feature vectors before passing them through a custom dense classification head.
- **System Optimization:** Perform systematic hyperparameter tuning adjusting learning rates, dropout configurations, and batch sizes to improve the cross-domain generalization gap.

4. Datasets and Scope

The project utilizes 3,000 labeled review sentences sourced from *Kotzias et al., KDD 2015*. The analysis satisfies the requirement for multiple datasets by crossing boundaries between three distinct commercial sectors:

- **Domain A (Source):** Amazon Product/Electronics Reviews (Tabular text data).
- **Domain B (Target 1):** IMDb Movie and Entertainment Reviews (Tabular text data).
- **Domain C (Target 2):** Yelp Restaurant and Hospitality Reviews (Tabular text data).

5. Primary Goal and Target Deliverable

Target: The final deliverable is an interactive, production-ready web application that demonstrates the real-world viability of the models.
End-User Experience: A business user will input a review sentence, select the domain origin of the text, and submit. The application will serve the predicted sentiment (positive/negative), output a visualization of the probability distribution, and display a domain-drift confidence score, proving the model's operational utility across industries.

6. Key Performance Indicators (KPIs) and Definition of Success

Because accuracy alone is insufficient for evaluating imbalanced real-world applications, success is defined by achieving the following specific engineering KPIs:

- **KPI 1: Source Domain Accuracy:** The final optimized hybrid model must achieve a baseline accuracy and Macro F1-score of >85% on the original training domain data.
- **KPI 2: Target Domain Retention (Cross-Domain Score):** The model must maintain a Macro F1-score of >75% when evaluated against the unseen target domains, demonstrating successful generalization rather than strict overfitting.
- **KPI 3: Inference Latency:** The deployed front-end application must generate a sentiment prediction and confidence score in under 2.0 seconds per request, satisfying the real-time operational requirements of a digital product dashboard.
- **KPI 4: Pipeline Reliability:** The underlying GitHub repository and test suite must pass automated quality gates (Unit testing via pytest) proving robust data handling of null values and proper sequence padding prior to tokenization